Last Time

Test of Proportions

T-tests of One Mean

Estimating σ^2

Today

T-tests of One Mean

Estimating σ^2, revisited

Behrens-Fisher Problem

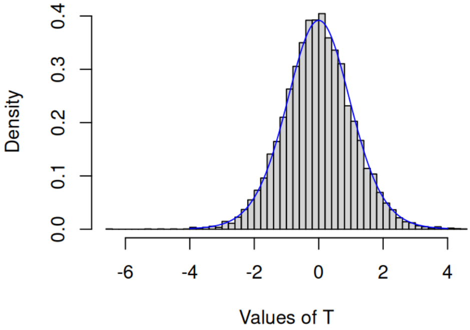
The *t*-distribution

t-table

```
bignum <- 10000 # number of simulated experiments
N <- 15 # sample size
pop.var <- 93 # population variance
pop.mean <- 47 # population mean

s2 <- S2 <- t.stat <- x.bar <- vector()

# Sample from a normal(pop.mean,pop.variance) distribution bignum times
# Compute the sample mean and variances and t-statistic
for (i in 1:bignum) {
  X <- rnorm(N,pop.mean,sqrt(pop.var))
  x.bar[i] <- mean(X)
  s2[i] <- var(X)
  S2[i] <- sum((X-x.bar[i])^2)/N # "Big S" squared, "population" variance
  t.stat[i] <- (x.bar[i] - pop.mean)/sqrt(s2[i]/N)
}

# What is the mean of the means?
cat("The population mean is ",pop.mean,
    " and the mean of the means is ",mean(x.bar))

# What is the mean of the variances?
cat("The population variance is ",pop.var,
    " and the mean of the sample variances is ",mean(s2))


# What is the mean of the "population" variances
cat("The population variance is ",pop.var,
    " and the mean of the "population" variances is ",mean(S2))
```

The *t*-distribution

```
> # What is the mean of
the means?
The population mean is  47
and the mean of the means is
46.95188
>
> # What is the mean of
the variances?
The population variance is
93  and the mean of the sample
variances is  92.42257
>
> # What is the mean of
the "population" variances
The population variance is
93  and the mean of the
'population' variances is
86.26107
```



The Sample Variance

Given a sample $\{X_1, X_2, \ldots, X_N\}$, we don't know σ^2, but we can compute

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(X_i - \overline{X})^2.$$

We showed that s^2 a good estimate of σ^2 by simulation.

$E[s^2] = \mu_{s^2} = \sigma^2 \Rightarrow s^2$ is *unbiased*, and therefore a good point estimate of σ^2 is s^2.

This means that $\hat{\sigma}_{\overline{X}} = s_{\overline{X}} = \frac{s}{\sqrt{N}}$, is a good estimate of the standard error of $\overline{X}$.

The (One-Sample) t-Test

Define: Test statistic

$$T_{\overline{X}} = \frac{\overline{X} - \mu}{s_{\overline{X}}},$$

where $s_{\overline{X}}$ is an estimate of $\sigma_{\overline{X}}$.

Two random variables (remember that any statistic is a random variable)

- $\overline{X} \sim \mathcal{N}(\mu, \sigma^2/N)$
- $s_{\overline{X}} \sim$ something else

- $T_{\overline{X}} \sim t(N-1)$, kind of like a z-score but not quite.

GSS Example

Is the age of the GSS respondents significantly older than 30?

- One-tailed test:

$$\begin{aligned} H_0 &: \quad \mu \leq 30 \\ H_1 &: \quad \mu > 30 \end{aligned}$$

- Test the null hypothesis at with a Type I error rate α equal to 0.05. (Always pick your α level before you do the test.)

GSS Example

-
$$\begin{aligned} \overline{X} &= 39.2462 \\ s^2 &= 196.1881 \\ N &= 873 \end{aligned}$$

-
$$T = \frac{\overline{X} - \mu_0}{s/\sqrt{N}} = \frac{39.2462 - 30}{14.0067/29.5466} = 19.5046$$

- $T \sim t(872) \approx \mathcal{N}(0, 1)$
- $Pr(T > 19.5046) = 0.0$ (p-value < 0.05)

T-tests in R

```
> t.test(data$AGE,alternative="greater",mu=30)

 One Sample t-test

data:  data$AGE
t = 19.505, df = 872, p-value < 2.2e-16
alternative hypothesis: true mean is greater than 30
95 percent confidence interval:
 38.4657      Inf
sample estimates:
mean of x
 39.24628
```

Don't Forget

Confidence intervals around a single mean:

$$\overline{X} \pm t_{\alpha/2}(N-1)\frac{s}{\sqrt{N}}$$

What Can We Say About A Small p-Value?

Suppose $H_0 : \mu = 38$.

Test 1: $\overline{X} = 38.97$, $p = 0.02$

Test 2: $\overline{X} = 39.36$, $p = 0.002$ "more significant"

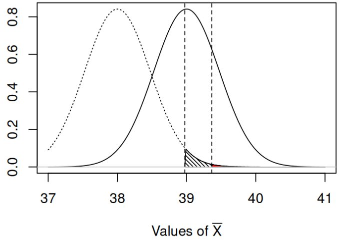
What Can We Say About A Small p-Value?

Suppose $H_0 : \mu = 38$.

Test 1: $\overline{X} = 38.97$, $p = 0.02$

Test 2: $\overline{X} = 39.36$, $p = 0.002$ "more significant"

Rejecting $H_0 : \mu = 38$ says nothing about the alternative mean. Let the "true" alternative mean be $\mu_1 = 39$. Then the "true" sampling distribution of $\overline{X} \sim \mathcal{N}(39, \sigma_{\overline{X}})$, where $\hat{\sigma}_{\overline{X}} = 0.4741$.



GSS Example

Can money buy happiness? Look at the income and happiness variables in the GSS dataset.

- Collapse the happiness rating (1=very happy, 2=pretty happy, 3=not too happy) into happy vs. not happy (two groups).
- If money buys some happiness, the income for the happy group should be higher than the income for the not happy group.

```
data <- read.csv('GSS_1980.csv')

# Collapse HAPPY variable and rename income
happy <- data$HAPPY < 3
money <- data$REALRINC
```

Income Statistics

	Happy	Unhappy
N	770	103
$\overline{X}$	\$23,849.19	\$17,924.87
s	\$24,780.02	\$17,665.58

What is H_0?

- Almost always: do the populations of the two samples have the same mean?
- $H_0 : \mu_1 = \mu_2$
- ...which is the same as $H_0 : \mu_1 - \mu_2 = 0$
- How will we estimate $\mu_1 - \mu_2$?

Sampling Distribution of $\overline{X}_1 - \overline{X}_2$

We know that $\overline{X} \sim \mathcal{N}(\mu, \sigma^2/N)$.
What is the sampling distribution of $\overline{X}_1 - \overline{X}_2$?
Assume:

- All $\overline{X}_i \sim \mathcal{N}(\mu_i, \sigma^2/N_i)$: different groups of different sizes have different means, but all groups have the same population variance σ^2.
- The groups are independent: different people are in each group and they are not connected to each other in any way.

Sampling Distribution of $\overline{X}_1 - \overline{X}_2$

- Both $\overline{X}_1$ and $\overline{X}_2$ follow normal distributions (CLT).
- If $\overline{X}_1 \sim \mathcal{N}(\mu_1, \sigma^2/N)$ and $\overline{X}_2 \sim \mathcal{N}(\mu_2, \sigma^2/N)$,
  - How is $\overline{X}_1 - \overline{X}_2$ distributed?
  - What is $E[\overline{X}_1 - \overline{X}_2]$?
  - What is $Var[\overline{X}_1 - \overline{X}_2]$?

Mean and Variance of the Difference

$$\begin{aligned} E[\overline{X}_1 - \overline{X}_2] &= E[\overline{X}_1] - E[\overline{X}_2] \\ &= \mu_1 - \mu_2. \end{aligned}$$

$$\begin{aligned} Var[\overline{X}_1 - \overline{X}_2] &= Var[\overline{X}_1] + Var[\overline{X}_2] \\ &= \sigma^2/N_1 + \sigma^2/N_2. \end{aligned}$$

Distribution of $\overline{X}_1$ and $\overline{X}_2$

Because both $\overline{X}_1$ and $\overline{X}_2$ are normally distributed, $\overline{X}_1 - \overline{X}_2$ is normally distributed. Therefore,

$$\overline{X}_1 - \overline{X}_2 \sim \mathcal{N}(\mu_1 - \mu_2, \sigma^2/N_1 + \sigma^2/N_2)$$

- $H_0 : \mu_1 = \mu_2 \rightarrow H_0 : \mu_1 - \mu_2 = 0$
- What to do about $\sigma^2/N_1 + \sigma^2/N_2$?
- (Note that $\sigma^2/N_1 + \sigma^2/N_2 = \sigma^2(1/N_1 + 1/N_2)$.)

Estimating the Standard Error

1. $\dfrac{\sigma^2}{N_1} + \dfrac{\sigma^2}{N_2} \Rightarrow \dfrac{s_1^2}{N_1} + \dfrac{s_2^2}{N_2}$

Estimating the Standard Error

1. $\dfrac{\sigma^2}{N_1} + \dfrac{\sigma^2}{N_2} \Rightarrow \dfrac{s_1^2}{N_1} + \dfrac{s_2^2}{N_2}$
2. Pool the variance: devise a single estimate of the common σ^2 using both samples

Pooling Variance

- Recenter measurements around 0 by subtracting off the mean
- Some observations:
  - We're estimating variance so we don't care where the center of the observations is, only that the two groups share the same center.
  - Adding constants to a variable doesn't change the variance, so we can move the observations anywhere.
  - The numerator of the sample variance is $\sum_{i=1}^{N}(X_i - \overline{X})^2$: subtracting $\overline{X}$ from all the observations recenters them around 0. (The mean of the deviations around $\overline{X}$ is 0.)
  - If the samples come from populations with the same variance we can pool all the shifted observations together and compute the variance using all the observations.

Pooling the Variance

For two samples X and Y,

$$s_1^2 = \frac{\sum_{i=1}^{N_1}(X_i - \overline{X})^2}{N_1 - 1} \qquad s_2^2 = \frac{\sum_{i=1}^{N_2}(Y_i - \overline{Y})^2}{N_s - 1}$$

Call the numerators the sums of squares for X and Y: SSX and SSY. Then

$$s_P^2 = \frac{SSX + SSY}{(N_1 - 1) + (N_2 - 1)}$$

Pooling the Variance

Another way to think about it:

$$s_P^2 = \frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{(N_1 - 1) + (N_2 - 2)}$$

A weighted average of the two sample variances.

$$\hat{\sigma}^2 = s_P^2$$

Income Statistics

	Happy	Unhappy
N	770	103
$\overline{X}$	\$23,849.19	\$17,924.87
s	\$24,780.02	\$17,665.58

GSS Example

Do happy people have higher incomes than unhappy people?

$$\begin{aligned} H_0 &: \quad \mu_H = \mu_U \\ H_1 &: \quad \mu_H > \mu_U \end{aligned}$$

- Assume $\sigma_H^2 = \sigma_U^2$.
- Assume normality of the income measurements.
- Choose $\alpha = 0.05$.
- Estimate σ^2.
- Compute the t-statistic for the mean difference.
- Compute the p-value of the t-statistic.

Independent Samples t-Test

$$
\begin{aligned}
\overline{X}_H - \overline{X}_U &= \$5,924.32 \\
s_P^2 &= \frac{(770-1)(614,049,163) + (103-1)(312,072,780)}{(770-1)+(103-1)} \\
&= 24,055.89
\end{aligned}
$$

$$
\begin{aligned}
\hat{\sigma}_{\overline{X}_H - \overline{X}_U} &= s_P\sqrt{\frac{1}{N_H} + \frac{1}{N_U}} = 2523.86 \\
t &= \frac{5924.32 - 0}{2523.86} = 2.3473 \\
p &= 0.0096
\end{aligned}
$$

Independent Samples *t*-Test

```
> t.test(money~happy,alternative="less",var.equal=TRUE)

 Two Sample t-test

data:  money by happy
t = -2.3473, df = 871, p-value = 0.009566
alternative hypothesis: true difference in means between group
FALSE and group TRUE is less than 0
95 percent confidence interval:
     -Inf -1768.53
sample estimates:
mean in group FALSE  mean in group TRUE
           17924.87            23849.19
```

What if $\sigma_1^2 \neq \sigma_2^2$?

- Homoskedasticity: $\sigma_1^2 = \sigma_2^2$
- Heteroskedisticity: $\sigma_1^2 \neq \sigma_2^2$

Behrens-Fisher Problem: $\sigma_1^2 \neq \sigma_2^2$

$$t = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

Satterthwaite Correction

$$t' = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \sim t(df')$$

where

$$df' = \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{\left(\frac{s_1^2}{N_1}\right)^2}{N_1 - 1} + \frac{\left(\frac{s_2^2}{N_2}\right)^2}{N_2 - 1}}$$

Welch's Two-Sample t-Test

$$df' = 161.278$$

$$\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}} = 1956.348$$

$$t' = \frac{5924.32 - 0}{1956.348} = 3.028$$

$$p = Pr(T > t' \mid df = df') = 0.0014$$

Welch's Independent Samples *t*-Test

```
> t.test(money~happy,alternative="less",var.equal=FALSE)

 Welch Two Sample t-test

data:  money by happy
t = -3.0283, df = 161.28, p-value = 0.001432
alternative hypothesis: true difference in means between group
FALSE and group TRUE is less than 0
95 percent confidence interval:
      -Inf -2687.825
sample estimates:
mean in group FALSE  mean in group TRUE
           17924.87            23849.19
```

Which to Use

Remember assumptions:

- Samples are drawn from normal populations
- Homoskedasticity: $\sigma_1^2 = \sigma_2^2$
- Equal sample sizes (oops)
- *t*-test is robust to moderate violations of normality and homoskedasticity; the Type I error rate doesn't get much bigger than the α level.
  - Sample sizes need to be comparable
  - Very unequal sample sizes should use Welch's
- R uses Welch's test by default, because it is now standard practice.
- If you're worried, you can perform one of a number of tests for equality of variances, but making this routine can inflate Type I error rates (Hayes & Cai, 2007).

Tests for Equality of Variances

- Levene's test (DescTools::LeveneTest(money~happy))
- Fligner-Killeen test (stats::fligner.test(money~happy)
- F ratio (stats::var.test(money~happy))
- Bartlett test (stats::bartlett.test(money~happy))